DIGITAL PLANT

Storytelling, Interaction Table and Development Plan

Storytelling: The Life of the Plant

The experience begins with a seed. As the viewer interacts with the plant and adjusts its environment, the seed germinates and the plant grows. New leaves and branches are generated according to rules. When the plant reaches maturity, a bud forms and opens into a flower. The flower later fades and produces seeds. Supportive conditions help the plant continue growing, while prolonged unfavourable conditions can slow growth and lead to withering. The cycle ends with the plant's death; a collected seed may provide the starting point for a new cycle.
This is a designed digital experience rather than a scientifically complete botanical simulation. Timing, environmental rules and visual responses will be tuned for clarity and for a short demonstration.

Table 1 — Lifecycle, Viewer Interaction and Plant Response

N	Life Cycle Stage	Player Action	Plant Response	What Needs to Be Implemented
1	Seed	Click on the seed; adjust light and water	Germination begins under suitable conditions	Seed model, click handling, environmental parameters, conditions for growth to begin
2	Germination	Water the seed; adjust lighting	A sprout appears and develops at a certain rate	Sprout appearance animation, growth rate control
3	Growth	Click on the leaves and stem; adjust light and water	New leaves and branches appear; their size, position, and direction change	Geometry generation, branching rules, shape dependence on actions
4	Bud	Maintain growth conditions; interact with the plant	A bud forms; it develops faster under favorable conditions	Bud model, conditions for transition to flowering, growth animation

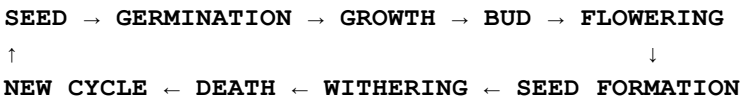
5	Flowering	Click on the bud or flower; continue to change conditions	The flower opens. Its shape may vary depending on the plant's development	Petal geometry, opening animation, shape variability
6	Seed formation	Interact with the flower or collect seeds	The flower wilts, and a seed pod forms	Transition between models, seed formation, interaction with harvesting
7	Withering	Adjust water and light while trying to sustain the plant	Under unfavorable conditions, growth slows down, and the leaves change color and position	State model, color change, growth slowdown, withering animation
8	Death and Renewal	Death and a New Beginning	Collect seeds and start a new cycle. The plant ends its life; a preserved seed can become the beginning of a new plant	Final state, seed preservation, new cycle

Table 2 — Main Interaction Systems

N	System	Purpose	Expected effect
1	Direct interaction	Let the viewer interact with leaves, stems, buds and seeds.	Clicks trigger meaningful responses, such as selecting a leaf, opening a flower or collecting seeds.
2	Water	Represent an environmental condition the viewer can change.	Water availability affects growth and condition; prolonged lack of water can contribute to withering.
3	Light	Represent a second environmental condition.	Light affects growth rate and may influence the direction of new growth.
4	Generative growth rules	Create variation while keeping the plant visually coherent.	Leaves and branches vary in number, size, position or angle according to parameters and viewer actions.
5	Plant condition	Connect environmental conditions to the lifecycle.	The plant can grow, remain stable, recover or wither depending on its simulated condition.

6	Seed collection / new cycle	Connect the ending to a possible beginning.	A collected seed can be used to initiate a new cycle.

Diagram 1 — Lifecycle and Interaction Logic



Interaction layer: direct clicks + environmental controls (water and light) → update growth parameters and plant condition → influence the current stage, growth rate, generated geometry and the possibility of withering. The lifecycle provides the narrative; interaction changes how the plant develops within that sequence.

Development Plan

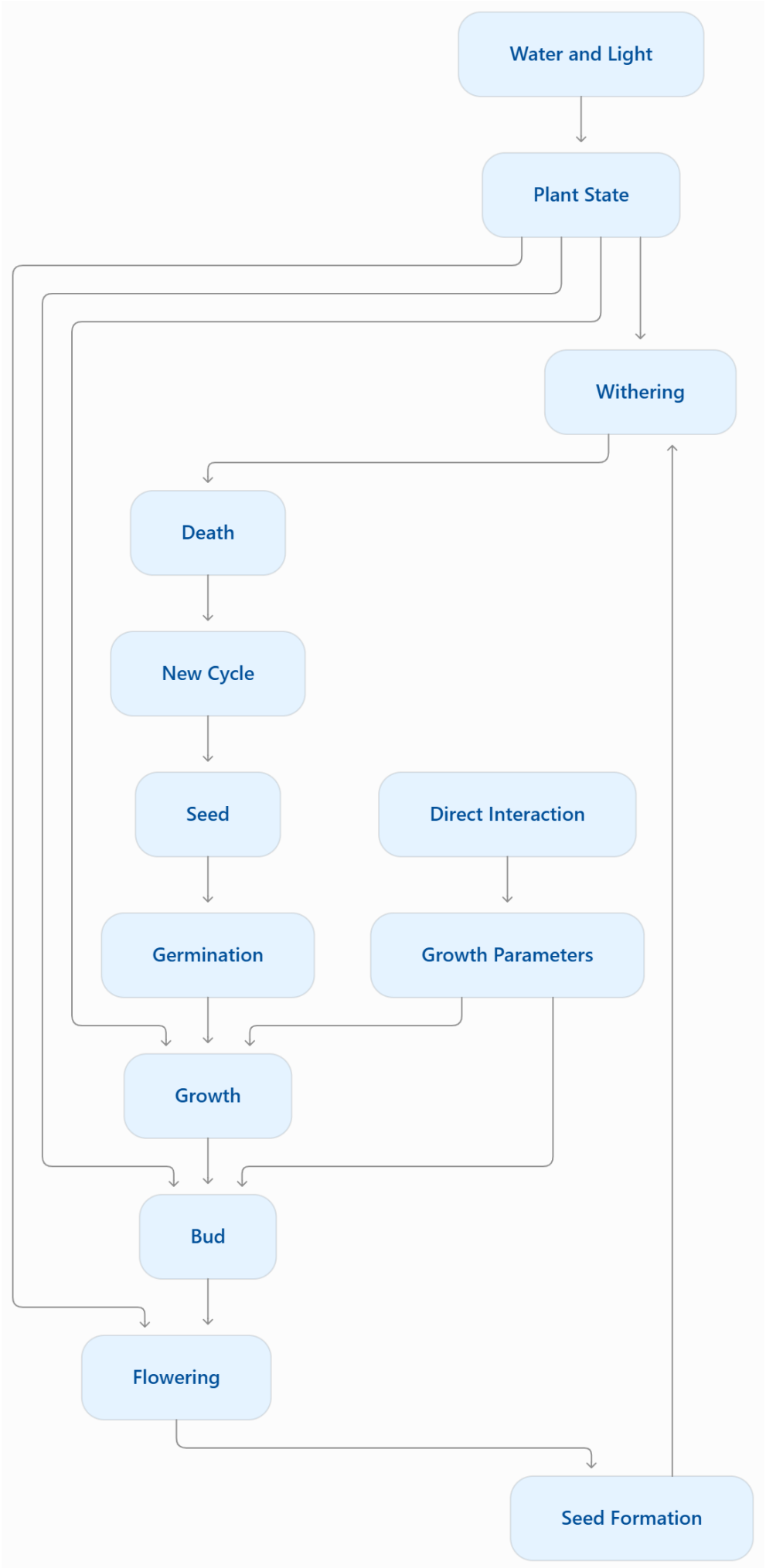
Phase 1 — Define the experience: Confirm lifecycle stages, viewer actions and expected responses. This document establishes the initial design.
Phase 2 — Stabilise the existing prototype: Use the current Three.js prototype with a stem, leaves, rotation and leaf-click interaction as the starting point.
Phase 3 — Add environmental controls: Introduce water and light values and connect them to simple growth and condition rules.
Phase 4 — Implement generative growth: Make leaf creation and placement respond to parameters so the plant can develop different forms.
Phase 5 — Build lifecycle stages: Add seed and germination first, then bud, flower, seed formation and withering incrementally.
Phase 6 — Connect the narrative: Make transitions between stages clear and ensure viewer actions produce visible consequences.
Phase 7 — Test and document: Test different interaction paths, record screenshots/video, and document what works and what needs further development.

Current Status and Scope

Current prototype work includes a 3D stem and leaves, plant rotation and click interaction with leaves. A leaf has also been explored as a separate model in Blender. The complete lifecycle, environmental controls, flowering, seed formation and withering are planned features and have not yet all been implemented.
Development will proceed incrementally. The first technical milestone is to connect simple water and light controls to growth and plant condition in the existing prototype. Further lifecycle stages will be added after this interaction is tested.

Intended Outcome

The intended outcome is a browser-based interactive generative plant experience. The viewer can interact with the plant, adjust environmental conditions, observe changes in growth and form, and follow the lifecycle from seed to death and possible renewal. The experience should communicate that the plant's appearance and condition are shaped by both generative rules and the viewer's choices.


Water and Light
Plant State
Withering
Death
New Cycle
Seed
Direct Interaction
Germination
Growth Parameters
Growth
Bud
Flowering
Seed Formation